SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

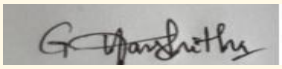
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA0301	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	CSE-AI (4th year)	**Date:**	05/10/2026

Code of Conduct

I G.Harshitha (192372384) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Topic given before the session and students speak on stage.
- Duration: 10 minutes (In class)

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Individual Task: Core Network Architecture in Cellular Systems

1. Introduction:

The **Core Network** is the central part of a cellular communication system. It connects mobile users to other users, the internet, and different communication services.

In simple words, the **Radio Access Network (RAN)** connects the mobile phone to the network, while the **Core Network manages and controls the communication**.

2. Basic Architecture:

A cellular network can be represented as:

Mobile Phone → Base Station → Core Network → Internet / Other Networks

The core network contains different components that perform important functions such as:

- User authentication
- Call management
- Mobility management
- Data routing
- Internet connectivity
- Charging and billing
- Security

3. Main Components:

1. Mobile Switching Center (MSC)

In traditional 2G/3G networks, the **MSC** manages voice calls and connects mobile users to other telephone networks.

2. Home Location Register (HLR)

The HLR stores important subscriber information such as the user's identity, services, and location-related information.

3. Visitor Location Register (VLR)

The VLR temporarily stores information about users currently present in a particular network area.

4. Serving GPRS Support Node (SGSN)

In 2G/3G packet networks, the SGSN manages mobile data communication and keeps track of the user's location.

5. Gateway GPRS Support Node (GGSN)

The GGSN connects the cellular network to external packet networks such as the Internet.

4. 4G Core Network

In **4G LTE**, the core network is called the **Evolved Packet Core (EPC)**.

Its major components are:

5. 5G Core Network

The **5G Core (5GC)** uses a more flexible **Service-Based Architecture**.

Important components include:

- **AMF:** Handles access and mobility management.
- **SMF:** Manages sessions and IP address allocation.
- **UPF:** Handles user data traffic.
- **UDM:** Manages subscriber information.
- **AUSF:** Performs authentication.
- **PCF:** Controls network policies.
- **NRF:** Helps network functions discover each other.

4. Conclusion:

To conclude, the **Core Network is the backbone of a cellular communication system**. It performs authentication, mobility management, session management, routing, security, and connectivity to external networks. From **2G and 3G core networks to 4G EPC and 5G Core**, the architecture has evolved to provide **higher speed, better security, lower latency, and support for a large number of connected devices**.

5.Geotag Photo:

